Assignment5

April 9, 2026

#Problem 2

a. Install the required libraries (e.g., transformers, torch).

b. Load a pre-trained Transformer model (e.g., bert-base-uncased) using the Hugging Face Transformers library.

c. Load the corresponding tokenizer.

d. Define an example sentence and tokenize it using the tokenizer.

e. Pass the tokenized input through the model and extract the attention weights.

[ ]:
```python
import torch
from transformers import BertModel, BertTokenizer

# Define the model name you want to load
model_name = 'bert-base-uncased'

print(f"Downloading and loading {model_name}...")

# Load the pre-trained model
model = BertModel.from_pretrained(model_name, attn_implementation="eager")

print("Successfully loaded the model!")
```

```
Downloading and loading bert-base-uncased…

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: bert-base-uncased
Key                                          | Status     |  |
---------------------------------------------+------------+--+-
cls.seq_relationship.weight                  | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight   | UNEXPECTED |  |
cls.seq_relationship.bias                    | UNEXPECTED |  |
cls.predictions.bias                         | UNEXPECTED |  |
cls.predictions.transform.dense.weight       | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |  |
cls.predictions.transform.dense.bias         | UNEXPECTED |  |

Notes:
```

```
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.
```

```
Successfully loaded the model!
```

[ ]:
```python
# Load the pre-trained tokenizer
tokenizer = BertTokenizer.from_pretrained(model_name)

print("Successfully loaded the tokenizer!")
```

```
Successfully loaded the tokenizer!
```

[ ]:
```python
# Define an example sentence
example_sentence = "Transformers have revolutionized natural language␣
↪processing!"

# 1. Break the sentence down into text tokens (sub-words)
text_tokens = tokenizer.tokenize(example_sentence)

# 2. Convert the sentence into the numerical format the model requires
# return_tensors='pt' returns PyTorch tensors
inputs = tokenizer(example_sentence, return_tensors='pt')

print("Original sentence:", example_sentence)
print("Text tokens:", text_tokens)
print("\nModel Inputs (Token IDs, Attention Mask, etc.):")
print(inputs)
```

```
Original sentence: Transformers have revolutionized natural language processing!
Text tokens: ['transformers', 'have', 'revolution', '##ized', 'natural',
'language', 'processing', '!']

Model Inputs (Token IDs, Attention Mask, etc.):
{'input_ids': tensor([[  101, 19081,  2031,  4329,  3550,  3019,  2653,  6364,
999,   102]]), 'token_type_ids': tensor([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]]),
'attention_mask': tensor([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])}
```

[ ]:
```python
inputs
```

[ ]:
```
{'input_ids': tensor([[  101, 19081,  2031,  4329,  3550,  3019,  2653,  6364,
999,   102]]), 'token_type_ids': tensor([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]]),
'attention_mask': tensor([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])}
```

[ ]:
```python
text_tokens
```

[ ]:
```
['transformers',
 'have',
 'revolution',
```

```
 '##ized',
 'natural',
 'language',
 'processing',
 '!']
```

[ ]:
```python
# Use torch.no_grad() because we are only evaluating/inferencing, not training
with torch.no_grad():
    # Pass the inputs to the model. The ** syntax unpacks the dictionary␣
 ↪automatically.
    outputs = model(**inputs, output_attentions=True)
```

##Problem 3

a. Select a specific layer and attention head from the model.

b. Visualize the attention weights using a heatmap or similar visualization.

c. Interpret the visualization and explain how the model distributes attention across tokens in the sentence.

[ ]:
```python
# Choose a specific layer (0 to 11 for BERT-base)
layer_idx = 0

# Choose a specific attention head (0 to 11 for BERT-base)
head_idx = 0

# Extract the attention weights for this specific layer
# `attention_weights` is a tuple of 12 tensors (one for each layer)
layer_attention = outputs.attentions[layer_idx]

# Select the specific head for the first item in our batch (index 0)
# Remember the shape is: (batch_size, num_heads, sequence_length,␣
 ↪sequence_length)
head_attention = layer_attention[0, head_idx, :, :]

print(f"Selected Layer: {layer_idx + 1} (Index {layer_idx})")
print(f"Selected Head: {head_idx + 1} (Index {head_idx})")
print(f"Shape of the selected attention matrix: {head_attention.shape}")
print("\nHere is a preview of the raw 10x10 attention matrix:")
print(head_attention)
```

```
Selected Layer: 1 (Index 0)
Selected Head: 1 (Index 0)
Shape of the selected attention matrix: torch.Size([10, 10])

Here is a preview of the raw 10x10 attention matrix:
tensor([[0.0762, 0.0634, 0.0760, 0.0683, 0.0540, 0.0538, 0.0830, 0.0821, 0.0558,
         0.3875],
```

```
        [0.0385, 0.0808, 0.0664, 0.1433, 0.0555, 0.0892, 0.0579, 0.1104, 0.2337,
         0.1243],
        [0.1308, 0.0899, 0.1425, 0.1268, 0.0474, 0.0669, 0.0350, 0.1117, 0.1514,
         0.0976],
        [0.0372, 0.1540, 0.0550, 0.0968, 0.0915, 0.1116, 0.0526, 0.1445, 0.2225,
         0.0342],
        [0.0395, 0.1385, 0.0616, 0.2308, 0.0395, 0.0766, 0.0647, 0.1022, 0.1432,
         0.1033],
        [0.0207, 0.1260, 0.0600, 0.1778, 0.0518, 0.0469, 0.1199, 0.2101, 0.1457,
         0.0412],
        [0.0347, 0.1707, 0.0257, 0.2530, 0.0458, 0.1096, 0.0387, 0.1823, 0.1156,
         0.0240],
        [0.0185, 0.1148, 0.0696, 0.2516, 0.0605, 0.0517, 0.1336, 0.0462, 0.1488,
         0.1047],
        [0.0477, 0.1204, 0.1076, 0.1477, 0.0580, 0.1132, 0.0482, 0.0857, 0.0867,
         0.1847],
        [0.0992, 0.0586, 0.0776, 0.0995, 0.0738, 0.0814, 0.0463, 0.1258, 0.1429,
         0.1949]])
```

[ ]:
```python
text_tokens
```

[ ]:
```
['transformers',
 'have',
 'revolution',
 '##ized',
 'natural',
 'language',
 'processing',
 '!']
```

[ ]:
```python
import matplotlib.pyplot as plt
import seaborn as sns

# Convert the PyTorch tensor to a standard NumPy array for plotting
attention_matrix = head_attention.numpy()

# Extract the exact tokens (including [CLS] and [SEP]) directly from the inputs
# This guarantees our labels match the 10x10 matrix perfectly
actual_tokens = tokenizer.convert_ids_to_tokens(inputs["input_ids"][0])

# Set up a 'fancy' matplotlib figure with a white background style
plt.figure(figsize=(10, 8))
sns.set_theme(style="white")

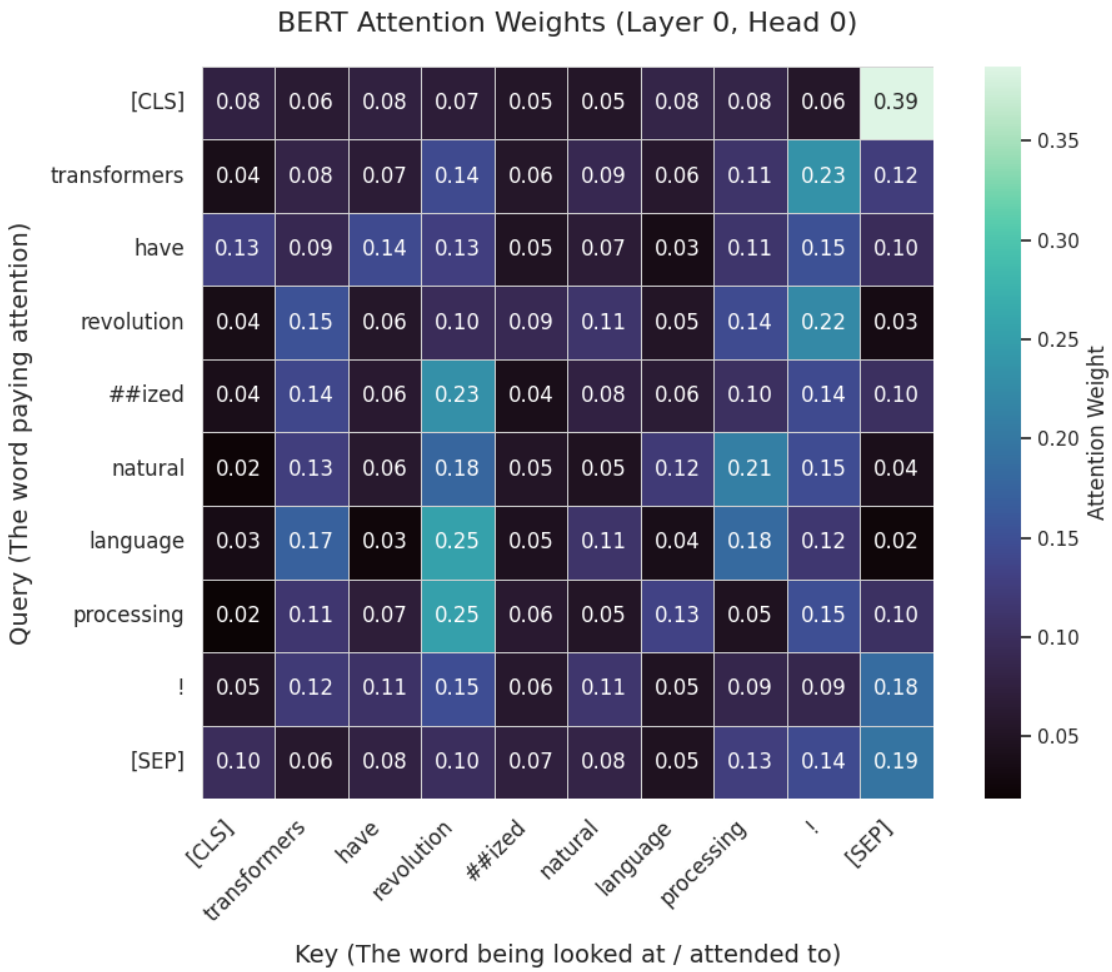
# Create a beautiful heatmap using Seaborn
ax = sns.heatmap(
    attention_matrix,
```

```
    annot=True,          # Show the actual numerical attention weights in the␣
 ↪boxes
    fmt=".2f",           # Format the numbers to 2 decimal places
    cmap="mako",         # Use a modern, high-contrast color map ('mako',␣
 ↪'magma', or 'viridis')
    cbar_kws={'label': 'Attention Weight'}, # Label the color bar
    square=True,         # Force the boxes to be perfect squares
    linewidths=0.5,      # Add a thin line between boxes for readability
    linecolor='lightgray'
)

# Set the labels to be the actual 10 tokens extracted above
ax.set_xticklabels(actual_tokens, rotation=45, ha='right', fontsize=12)
ax.set_yticklabels(actual_tokens, rotation=0, fontsize=12)

# Add descriptive titles and axis labels
plt.title(f"BERT Attention Weights (Layer {layer_idx}, Head {head_idx})",␣
 ↪fontsize=16, pad=20)
plt.xlabel("Key (The word being looked at / attended to)", fontsize=14,␣
 ↪labelpad=10)
plt.ylabel("Query (The word paying attention)", fontsize=14, labelpad=10)

# Adjust layout to make sure labels aren't cut off and display the plot
plt.tight_layout()
plt.show()
```



[ ]:
```
inputs
```

[ ]:
```
{'input_ids': tensor([[  101, 19081,  2031,  4329,  3550,  3019,  2653,  6364,
999,   102]]), 'token_type_ids': tensor([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]]),
'attention_mask': tensor([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])}
```

##3.c In its early layers, the model distributes its attention primarily toward foundational structure and basic identity. Most words place a heavy focus on themselves to maintain their core meaning, while also leaning on the special [CLS] and [SEP] boundary tokens. These special tokens act as neutral "parking spots" for attention when a word doesn't yet need deep context from the rest of the sentence.

At the same time, the model heavily prioritizes local grammar and word fragments. You will see strong attention weights between words that are right next to each other, as well as intense connections between sub-words (like "revolution" and "##ized"), as the model mathematically glues these pieces back together to understand them as a single concept

#4

##4.a. Modify the input sentence and repeat the analysis.

[ ]:
```
# 1. Define a completely new sentence (A classic NLP coreference test)
new_sentence = "The animal didn't cross the street because it was too tired."

# 2. Tokenize the new sentence and get the exact string tokens for our labels
new_inputs = tokenizer(new_sentence, return_tensors='pt')
new_tokens = tokenizer.convert_ids_to_tokens(new_inputs["input_ids"][0])

print(f"New Sentence: {new_sentence}")
print(f"New Tokens: {new_tokens}\n")

# 3. Pass the new sentence through the model
with torch.no_grad():
    new_outputs = model(**new_inputs, output_attentions=True)

# 4. Extract the attention weights for Layer 0, Head 0
# Indexing breakdown: attentions[layer][batch, head, query_seq, key_seq]
new_attention_matrix = new_outputs.attentions[0][0, 0, :, :].numpy()

# 5. Visualize the new attention weights!
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(14, 11))
sns.set_theme(style="white")

ax = sns.heatmap(
    new_attention_matrix,
    annot=True,
    fmt=".2f",
    cmap="mako",
    cbar_kws={'label': 'Attention Weight'},
    square=True,
    linewidths=0.5,
    linecolor='lightgray'
)

# Use our new_tokens for the X and Y labels
ax.set_xticklabels(new_tokens, rotation=45, ha='right', fontsize=12)
ax.set_yticklabels(new_tokens, rotation=0, fontsize=12)

plt.title("BERT Attention Weights (New Sentence | Layer 0, Head 0)",␣
 ↪fontsize=16, pad=20)
plt.xlabel("Key (The word being looked at)", fontsize=14, labelpad=10)
plt.ylabel("Query (The word paying attention)", fontsize=14, labelpad=10)
```

```
plt.tight_layout()
plt.show()
```

```
New Sentence: The animal didn't cross the street because it was too tired.
New Tokens: ['[CLS]', 'the', 'animal', 'didn', "'", 't', 'cross', 'the',
'street', 'because', 'it', 'was', 'too', 'tired', '.', '[SEP]']
```

[ ]:

```

```

#4.b. Compare how the attention patterns change with different inputs.

The "It" Pronoun (Crucial Observation): Look at the row for the word it (the Query). It distributes its attention fairly evenly, with its highest weights on itself (0.10), [CLS] (0.09), and punctuation. It pays only 0.05 attention to both animal and street. The model hasn't figured out the context yet!

Comparison

Sub-words vs Contractions: The previous heatmap showed how the model handles complex suffixes (gluing revolution to ##ized). This new one shows how it handles punctuation-based contractions.

The Diagonal: The previous heatmap likely had a stronger, more obvious diagonal line (words looking strictly at themselves to maintain their identity). In this specific sentence, Head 0 is much more active in looking across the sentence at verbs and nouns to map out the grammar.

Both heatmaps prove the exact same fundamental rule of Transformers: Early layers (like Layer 0) only care about basic syntax and structure. They park their attention on [CLS], punctuation, and adjacent grammar blocks.

0.1 4.C examine multiple layers and attention heads.

[ ]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Define the specific layers and heads we want to compare
# Format: (Layer_Index, Head_Index)
heads_to_examine = [
    (0, 0),   # Early layer, Head 0 (Syntax/Grammar)
    (0, 1),   # Early layer, Head 1 (Syntax/Grammar)
    (11, 0),  # Final layer, Head 0 (Deep Semantics)
    (11, 1)   # Final layer, Head 1 (Deep Semantics)
]

# Create a 2x2 grid of subplots
fig, axes = plt.subplots(4, 1, figsize=(36, 30))
axes = axes.flatten()

sns.set_theme(style="white")

for i, (l_idx, h_idx) in enumerate(heads_to_examine):
    # Extract the specific attention matrix
    matrix = new_outputs.attentions[l_idx][0, h_idx, :, :].numpy()

    # Plot the heatmap on the corresponding subplot
    sns.heatmap(
        matrix,
        annot=True,
        fmt=".2f",
        cmap="mako",
        cbar=False,          # Hide individual color bars to save space
        square=True,
        linewidths=0.5,
        linecolor='lightgray',
        ax=axes[i],
```

```
        annot_kws={"size": 8} # Make text slightly smaller to fit the grid
    )

    # Set the labels
    axes[i].set_xticklabels(new_tokens, rotation=45, ha='right', fontsize=10)
    axes[i].set_yticklabels(new_tokens, rotation=0, fontsize=10)
    axes[i].set_title(f"Layer {l_idx}, Head {h_idx}", fontsize=16, pad=10)

plt.tight_layout()
plt.show()
```

Layer 0, Head 0

	[CLS]	the	animal	didn	'	t	cross	the	street	because	it	was	too	tired	.	[SEP]
[CLS]	0.04	0.10	0.03	0.02	0.03	0.04	0.04	0.11	0.03	0.05	0.08	0.04	0.07	0.03	0.09	0.22
the	0.07	0.07	0.05	0.04	0.07	0.02	0.05	0.09	0.05	0.07	0.07	0.07	0.07	0.06	0.08	0.07
animal	0.03	0.02	0.06	0.13	0.07	0.04	0.04	0.02	0.15	0.04	0.04	0.05	0.06	0.14	0.06	0.05
didn	0.09	0.03	0.05	0.03	0.08	0.01	0.07	0.03	0.06	0.08	0.06	0.16	0.08	0.05	0.06	0.05
'	0.06	0.04	0.06	0.05	0.04	0.05	0.05	0.05	0.07	0.09	0.05	0.10	0.06	0.05	0.09	0.10
t	0.11	0.04	0.05	0.06	0.09	0.01	0.05	0.03	0.08	0.07	0.06	0.09	0.04	0.07	0.07	0.08
cross	0.04	0.03	0.11	0.11	0.08	0.06	0.03	0.03	0.15	0.04	0.06	0.05	0.05	0.07	0.04	0.05
the	0.06	0.08	0.05	0.04	0.08	0.03	0.05	0.09	0.04	0.07	0.08	0.07	0.07	0.05	0.08	0.05
street	0.04	0.04	0.12	0.11	0.06	0.09	0.03	0.04	0.06	0.04	0.05	0.05	0.05	0.06	0.05	0.10
because	0.08	0.05	0.04	0.05	0.07	0.02	0.04	0.05	0.03	0.05	0.08	0.12	0.05	0.04	0.16	0.09
it	0.09	0.03	0.05	0.05	0.06	0.05	0.05	0.04	0.05	0.10	0.04	0.07	0.06	0.09	0.08	0.08
was	0.10	0.03	0.06	0.05	0.11	0.03	0.04	0.04	0.04	0.07	0.04	0.16	0.06	0.06	0.08	0.04
too	0.07	0.02	0.04	0.06	0.04	0.10	0.05	0.02	0.04	0.14	0.05	0.06	0.03	0.06	0.12	0.09
tired	0.04	0.04	0.07	0.04	0.08	0.03	0.05	0.04	0.04	0.09	0.08	0.08	0.06	0.04	0.14	0.09
.	0.05	0.06	0.06	0.04	0.06	0.05	0.05	0.07	0.05	0.10	0.06	0.06	0.06	0.06	0.10	0.06
[SEP]	0.06	0.08	0.04	0.03	0.04	0.04	0.04	0.10	0.04	0.09	0.07	0.05	0.04	0.04	0.13	0.10

Layer 0, Head 1

	[CLS]	the	animal	didn	'	t	cross	the	street	because	it	was	too	tired	.	[SEP]
[CLS]	0.65	0.02	0.01	0.00	0.21	0.00	0.01	0.02	0.01	0.01	0.01	0.01	0.01	0.00	0.03	0.01
the	0.04	0.05	0.03	0.05	0.05	0.05	0.03	0.05	0.03	0.14	0.08	0.21	0.06	0.05	0.07	0.03
animal	0.06	0.02	0.08	0.07	0.02	0.08	0.09	0.03	0.12	0.04	0.01	0.01	0.02	0.20	0.01	0.14
didn	0.01	0.01	0.11	0.05	0.02	0.03	0.08	0.01	0.03	0.28	0.02	0.02	0.07	0.22	0.01	0.01
'	0.01	0.02	0.09	0.11	0.01	0.08	0.04	0.02	0.04	0.31	0.03	0.03	0.06	0.09	0.05	0.02
t	0.01	0.02	0.07	0.10	0.01	0.02	0.03	0.03	0.05	0.33	0.03	0.02	0.05	0.13	0.05	0.05
cross	0.04	0.06	0.11	0.03	0.01	0.02	0.04	0.06	0.30	0.01	0.03	0.01	0.01	0.13	0.02	0.11
the	0.03	0.05	0.04	0.05	0.05	0.05	0.03	0.05	0.04	0.14	0.08	0.17	0.05	0.06	0.06	0.04
street	0.02	0.02	0.27	0.03	0.03	0.08	0.06	0.02	0.12	0.02	0.02	0.02	0.05	0.10	0.02	0.12
because	0.00	0.02	0.09	0.15	0.01	0.04	0.05	0.02	0.07	0.16	0.05	0.10	0.08	0.14	0.02	0.01
it	0.00	0.01	0.13	0.09	0.02	0.08	0.06	0.01	0.15	0.09	0.01	0.02	0.05	0.24	0.01	0.04
was	0.00	0.01	0.12	0.16	0.01	0.05	0.04	0.01	0.11	0.12	0.02	0.02	0.04	0.25	0.01	0.02
too	0.01	0.02	0.13	0.09	0.01	0.04	0.04	0.02	0.03	0.38	0.03	0.03	0.02	0.12	0.01	0.02
tired	0.01	0.01	0.15	0.07	0.03	0.07	0.12	0.01	0.09	0.07	0.04	0.08	0.08	0.07	0.02	0.07
.	0.00	0.06	0.01	0.03	0.01	0.02	0.01	0.06	0.01	0.15	0.20	0.24	0.03	0.01	0.17	0.01
[SEP]	0.02	0.06	0.02	0.01	0.07	0.02	0.01	0.06	0.02	0.12	0.12	0.10	0.03	0.01	0.29	0.02

Layer 11, Head 0

	[CLS]	the	animal	didn	'	t	cross	the	street	because	it	was	too	tired	.	[SEP]
[CLS]	0.16	0.04	0.13	0.02	0.19	0.01	0.02	0.01	0.01	0.04	0.04	0.01	0.00	0.03	0.18	0.12
the	0.06	0.02	0.03	0.01	0.36	0.01	0.00	0.00	0.00	0.03	0.02	0.01	0.01	0.01	0.37	0.06
animal	0.02	0.01	0.02	0.00	0.43	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.01	0.44	0.03
didn	0.06	0.02	0.02	0.01	0.35	0.01	0.01	0.00	0.00	0.02	0.01	0.01	0.00	0.00	0.36	0.11
'	0.01	0.01	0.01	0.01	0.44	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.46	0.02
t	0.07	0.04	0.04	0.01	0.32	0.01	0.01	0.01	0.00	0.04	0.01	0.01	0.00	0.00	0.33	0.09
cross	0.07	0.01	0.01	0.00	0.39	0.00	0.01	0.00	0.01	0.01	0.01	0.00	0.00	0.01	0.41	0.07
the	0.07	0.03	0.02	0.01	0.33	0.01	0.01	0.00	0.00	0.03	0.02	0.01	0.00	0.02	0.33	0.10
street	0.02	0.01	0.02	0.01	0.43	0.00	0.01	0.00	0.01	0.01	0.01	0.00	0.00	0.01	0.44	0.03
because	0.11	0.04	0.03	0.02	0.23	0.01	0.02	0.03	0.01	0.05	0.01	0.01	0.01	0.00	0.23	0.20
it	0.08	0.02	0.02	0.01	0.35	0.01	0.00	0.00	0.00	0.02	0.01	0.01	0.00	0.01	0.37	0.08
was	0.04	0.01	0.01	0.00	0.43	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.45	0.05
too	0.04	0.00	0.01	0.01	0.43	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.44	0.03
tired	0.04	0.01	0.03	0.01	0.40	0.00	0.01	0.00	0.00	0.01	0.01	0.01	0.01	0.02	0.41	0.03
.	0.01	0.01	0.01	0.01	0.44	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.46	0.02
[SEP]	0.12	0.06	0.08	0.04	0.17	0.02	0.02	0.01	0.02	0.05	0.04	0.01	0.01	0.02	0.17	0.14

Layer 11, Head 1

	[CLS]	the	animal	didn	'	t	cross	the	street	because	it	was	too	tired	.	[SEP]
[CLS]	0.16	0.06	0.15	0.02	0.07	0.01	0.03	0.04	0.03	0.04	0.03	0.01	0.01	0.03	0.07	0.24
the	0.04	0.01	0.01	0.01	0.40	0.01	0.00	0.01	0.01	0.03	0.00	0.01	0.01	0.01	0.41	0.05
animal	0.07	0.02	0.01	0.02	0.31	0.02	0.02	0.03	0.01	0.02	0.01	0.01	0.01	0.01	0.32	0.11
didn	0.03	0.01	0.00	0.00	0.43	0.00	0.01	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.44	0.04
'	0.01	0.01	0.00	0.01	0.46	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.47	0.02
t	0.04	0.01	0.00	0.01	0.42	0.01	0.02	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.42	0.04
cross	0.03	0.01	0.00	0.00	0.43	0.01	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.43	0.06
the	0.04	0.02	0.01	0.01	0.39	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.40	0.11
street	0.05	0.04	0.01	0.01	0.31	0.01	0.03	0.03	0.01	0.01	0.01	0.01	0.00	0.01	0.31	0.15
because	0.06	0.04	0.01	0.04	0.32	0.04	0.01	0.01	0.01	0.02	0.01	0.01	0.01	0.01	0.32	0.06
it	0.02	0.00	0.00	0.00	0.45	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.46	0.03
was	0.01	0.00	0.00	0.00	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.49	0.01
too	0.01	0.00	0.00	0.00	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.49	0.02
tired	0.04	0.02	0.01	0.02	0.31	0.03	0.02	0.01	0.02	0.05	0.01	0.02	0.05	0.01	0.32	0.04
.	0.01	0.01	0.00	0.01	0.46	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.47	0.02
[SEP]	0.08	0.03	0.03	0.02	0.31	0.02	0.02	0.02	0.02	0.03	0.01	0.01	0.01	0.02	0.31	0.07

##d. Discuss whether deeper layers appear to capture different types of information.

Early Layers (Layer 0, Heads 0 & 1): In the top two heatmaps, the attention is relatively dispersed and highly localized. Words are primarily looking at their immediate neighbors, paying attention to themselves (the diagonal), or defaulting to the [CLS] token. The model is simply mapping out the foundational grammar and syntax of the sentence.

Deep Layers (Layer 11, Heads 0 & 1): In the bottom two heatmaps, the pattern changes completely into stark, intense vertical stripes. Almost every single word in the sentence is aggressively paying attention to just one or two specific tokens—in this case, punctuation like the apostrophe (') and the period (.).

The "Parking Lot" Phenomenon: This vertical banding is a famous and heavily researched behavior in BERT! By Layer 11, the model already understands the basic grammar. When an attention head doesn't have a specific semantic relationship it needs to focus on for a given word, it uses punctuation tokens as a "parking spot" (or a "no-op") to safely deposit attention. Furthermore, it uses these structural tokens as global aggregation hubs to summarize the broad meaning of the entire sentence, proving the deeper layers process text as a unified, global concept rather than a sequence of individual words.

#5

##5a. Summarize your findings from the attention analysis.

Hierarchical Processing: The BERT model processes language in a hierarchical manner, moving from simple structural mapping to complex semantic understanding.

Local to Global Shift: Early layers act as syntax analyzers, focusing heavily on local, adjacent words and basic grammar. Deeper layers act as semantic analyzers, abandoning local syntax to focus on broad context and resolving complex relationships (like ambiguous pronouns).

Structural Anchors: Deep layers frequently employ the "parking lot" phenomenon, utilizing structural tokens (like punctuation marks and the [CLS] token) as safe resting places for attention or as central hubs to aggregate the overarching meaning of the entire sequence.

##5.b. Discuss what the attention visualizations reveal about how Transformers process language.

These visualizations reveal that Transformers do not process language as a static sequence of dictionary definitions. Instead, they process language dynamically and contextually.

When a Transformer reads a word, it doesn't just look at the word itself; it uses the self-attention mechanism as a "routing system" to pull in mathematical context from everywhere else in the sentence. The visualizations show us exactly how this routing works in real-time. By distributing varying amounts of attention across different tokens, the model continuously updates the meaning of every word based on its surroundings. It proves that in a Transformer, the meaning of a word is defined entirely by the company it keeps.

##5.c Reflect on any limitations of attention-based interpretation.

While heatmaps are visually appealing and offer great intuition, interpreting a Transformer solely through its attention weights has significant scientific limitations:

"Attention is Not Explanation": There is an ongoing debate in NLP (highlighted by a famous 2019 paper titled "Attention is not Explanation") arguing that a high attention weight does not strictly equal causal importance. Just because word A pays heavy attention to word B does not guarantee that word B was the primary reason the model made a specific final prediction.

Ignoring the Feed-Forward Networks: Attention is only half of the architecture. After the attention weights route the information, the data is passed through complex Feed-Forward Neural Networks and Residual Connections (shortcuts). These non-linear layers can drastically alter, suppress, or transform the information, which the attention heatmap completely fails to show.

The Multi-Head Complexity: We only looked at a few individual heads. A standard BERT model has 144 total attention heads (12 layers $\times$ 12 heads). Focusing on a single 10x10 grid isolates just a fraction of a percent of the model's actual "thinking." It is incredibly difficult for humans to grasp the combined, high-dimensional reality of 144 heads working together simultaneously.

[1]:
```
!apt-get install texlive-xetex texlive-fonts-recommended texlive-plain-generic␣
↪-y
!jupyter nbconvert --to pdf /content/Assignment5.ipynb
```

```
Reading package lists… Done
Building dependency tree… Done
Reading state information… Done
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries texlive-latex-base
  texlive-latex-extra texlive-latex-recommended texlive-pictures tipa
  xfonts-encodings xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
  fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
  | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
  texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
  icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
  texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
  texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
  default-jre-headless tipa-doc
The following NEW packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
```

```
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-extra
  texlive-latex-recommended texlive-pictures texlive-plain-generic
  texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 53 newly installed, 0 to remove and 2 not upgraded.
Need to get 182 MB of archives.
After this operation, 571 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
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0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.3 [60.6 kB]
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1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
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2.004.5-6.1 [4,532 kB]
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20201225-1build1 [397 kB]
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20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
```

```
java all 1.2-2 [60.3 kB]
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amd64 2021.20210626.59705-1ubuntu0.3 [39.1 kB]
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all 1.18 [5,336 B]
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3.0.2-7ubuntu2.12 [50.1 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 http://archive.ubuntu.com/ubuntu jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
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all 1.7.0-3ubuntu0.2 [52.5 kB]
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0.3.2-1ubuntu0.1 [24.9 kB]
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amd64 3.0.2-7ubuntu2.12 [5,113 kB]
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amd64 2021.20210626.59705-1ubuntu0.3 [55.8 kB]
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2.004.5-6.1 [9,471 kB]
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all 12.2-1ubuntu1 [185 kB]
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1.41-4build2 [61.3 kB]
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20180621-3.1 [6,209 kB]
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```

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all 2021.20220204-1 [1,128 kB]
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1:1.8.16-2 [207 kB]
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recommended all 2021.20220204-1 [14.4 MB]
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all 2021.20220204-1 [13.9 MB]
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generic all 2021.20220204-1 [27.5 MB]
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[2,967 kB]
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2021.20220204-1 [12.4 MB]
Fetched 182 MB in 5s (33.5 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages …
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(Reading database … 118194 files and directories currently installed.)
Preparing to unpack …/00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
…
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) …
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Preparing to unpack …/01-fonts-lato_2.0-2.1_all.deb …
Unpacking fonts-lato (2.0-2.1) …
Selecting previously unselected package poppler-data.
Preparing to unpack …/02-poppler-data_0.4.11-1_all.deb …
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Selecting previously unselected package tex-common.
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Unpacking tex-common (6.17) …
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Preparing to unpack …/04-fonts-urw-base35_20200910-1_all.deb …
Unpacking fonts-urw-base35 (20200910-1) …
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Selecting previously unselected package libidn12:amd64.
Preparing to unpack …/06-libidn12_1.38-4ubuntu1_amd64.deb …
Unpacking libidn12:amd64 (1.38-4ubuntu1) …
Selecting previously unselected package libijs-0.35:amd64.
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Unpacking libijs-0.35:amd64 (0.35-15build2) …
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Preparing to unpack …/08-libjbig2dec0_0.19-3build2_amd64.deb …
Unpacking libjbig2dec0:amd64 (0.19-3build2) …
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Preparing to unpack …/09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb …
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack …/10-libkpathsea6_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack …/11-libwoff1_1.0.2-1build4_amd64.deb …
Unpacking libwoff1:amd64 (1.0.2-1build4) …
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Unpacking dvisvgm (2.13.1-1) …
Selecting previously unselected package fonts-lmodern.
Preparing to unpack …/13-fonts-lmodern_2.004.5-6.1_all.deb …
Unpacking fonts-lmodern (2.004.5-6.1) …
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Unpacking fonts-noto-mono (20201225-1build1) …
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Selecting previously unselected package libcommons-parent-java.
Preparing to unpack …/17-libcommons-parent-java_43-1_all.deb …
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Preparing to unpack …/18-libcommons-logging-java_1.2-2_all.deb …
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Selecting previously unselected package libptexenc1:amd64.
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…
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package rubygems-integration.
Preparing to unpack …/20-rubygems-integration_1.18_all.deb …
Unpacking rubygems-integration (1.18) …
Selecting previously unselected package ruby3.0.
Preparing to unpack …/21-ruby3.0_3.0.2-7ubuntu2.12_amd64.deb …
Unpacking ruby3.0 (3.0.2-7ubuntu2.12) …
Selecting previously unselected package ruby-rubygems.
Preparing to unpack …/22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb …
```

```
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) …
Selecting previously unselected package ruby.
Preparing to unpack …/23-ruby_1%3a3.0~exp1_amd64.deb …
Unpacking ruby (1:3.0~exp1) …
Selecting previously unselected package rake.
Preparing to unpack …/24-rake_13.0.6-2_all.deb …
Unpacking rake (13.0.6-2) …
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack …/25-ruby-net-telnet_0.1.1-2_all.deb …
Unpacking ruby-net-telnet (0.1.1-2) …
Selecting previously unselected package ruby-webrick.
Preparing to unpack …/26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb …
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) …
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack …/27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb …
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack …/28-libruby3.0_3.0.2-7ubuntu2.12_amd64.deb …
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.12) …
Selecting previously unselected package libsynctex2:amd64.
Preparing to unpack …/29-libsynctex2_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack …/30-libteckit0_2.5.11+ds1-1_amd64.deb …
Unpacking libteckit0:amd64 (2.5.11+ds1-1) …
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack …/31-libtexlua53_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
…/32-libtexluajit2_2021.20210626.59705-1ubuntu0.3_amd64.deb …
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack …/33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb …
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Selecting previously unselected package xfonts-encodings.
Preparing to unpack …/34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb …
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) …
Selecting previously unselected package xfonts-utils.
Preparing to unpack …/35-xfonts-utils_1%3a7.7+6build2_amd64.deb …
Unpacking xfonts-utils (1:7.7+6build2) …
Selecting previously unselected package lmodern.
Preparing to unpack …/36-lmodern_2.004.5-6.1_all.deb …
Unpacking lmodern (2.004.5-6.1) …
Selecting previously unselected package preview-latex-style.
Preparing to unpack …/37-preview-latex-style_12.2-1ubuntu1_all.deb …
```

```
Unpacking preview-latex-style (12.2-1ubuntu1) …
Selecting previously unselected package t1utils.
Preparing to unpack …/38-t1utils_1.41-4build2_amd64.deb …
Unpacking t1utils (1.41-4build2) …
Selecting previously unselected package teckit.
Preparing to unpack …/39-teckit_2.5.11+ds1-1_amd64.deb …
Unpacking teckit (2.5.11+ds1-1) …
Selecting previously unselected package tex-gyre.
Preparing to unpack …/40-tex-gyre_20180621-3.1_all.deb …
Unpacking tex-gyre (20180621-3.1) …
Selecting previously unselected package texlive-binaries.
Preparing to unpack …/41-texlive-
binaries_2021.20210626.59705-1ubuntu0.3_amd64.deb …
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package texlive-base.
Preparing to unpack …/42-texlive-base_2021.20220204-1_all.deb …
Unpacking texlive-base (2021.20220204-1) …
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack …/43-texlive-fonts-recommended_2021.20220204-1_all.deb …
Unpacking texlive-fonts-recommended (2021.20220204-1) …
Selecting previously unselected package texlive-latex-base.
Preparing to unpack …/44-texlive-latex-base_2021.20220204-1_all.deb …
Unpacking texlive-latex-base (2021.20220204-1) …
Selecting previously unselected package libfontbox-java.
Preparing to unpack …/45-libfontbox-java_1%3a1.8.16-2_all.deb …
Unpacking libfontbox-java (1:1.8.16-2) …
Selecting previously unselected package libpdfbox-java.
Preparing to unpack …/46-libpdfbox-java_1%3a1.8.16-2_all.deb …
Unpacking libpdfbox-java (1:1.8.16-2) …
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack …/47-texlive-latex-recommended_2021.20220204-1_all.deb …
Unpacking texlive-latex-recommended (2021.20220204-1) …
Selecting previously unselected package texlive-pictures.
Preparing to unpack …/48-texlive-pictures_2021.20220204-1_all.deb …
Unpacking texlive-pictures (2021.20220204-1) …
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack …/49-texlive-latex-extra_2021.20220204-1_all.deb …
Unpacking texlive-latex-extra (2021.20220204-1) …
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack …/50-texlive-plain-generic_2021.20220204-1_all.deb …
Unpacking texlive-plain-generic (2021.20220204-1) …
Selecting previously unselected package tipa.
Preparing to unpack …/51-tipa_2%3a1.3-21_all.deb …
Unpacking tipa (2:1.3-21) …
Selecting previously unselected package texlive-xetex.
Preparing to unpack …/52-texlive-xetex_2021.20220204-1_all.deb …
Unpacking texlive-xetex (2021.20220204-1) …
Setting up fonts-lato (2.0-2.1) …
```

```
Setting up fonts-noto-mono (20201225-1build1) …
Setting up libwoff1:amd64 (1.0.2-1build4) …
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up libijs-0.35:amd64 (0.35-15build2) …
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up libfontbox-java (1:1.8.16-2) …
Setting up rubygems-integration (1.18) …
Setting up libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Setting up fonts-urw-base35 (20200910-1) …
Setting up poppler-data (0.4.11-1) …
Setting up tex-common (6.17) …
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) …
Setting up libteckit0:amd64 (2.5.11+ds1-1) …
Setting up libapache-pom-java (18-1) …
Setting up ruby-net-telnet (0.1.1-2) …
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) …
Setting up t1utils (1.41-4build2) …
Setting up libidn12:amd64 (1.38-4ubuntu1) …
Setting up fonts-texgyre (20180621-3.1) …
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up ruby-webrick (1.7.0-3ubuntu0.2) …
Setting up fonts-lmodern (2.004.5-6.1) …
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Setting up libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) …
Setting up teckit (2.5.11+ds1-1) …
Setting up libpdfbox-java (1:1.8.16-2) …
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Setting up preview-latex-style (12.2-1ubuntu1) …
Setting up libcommons-parent-java (43-1) …
Setting up dvisvgm (2.13.1-1) …
Setting up libcommons-logging-java (1.2-2) …
Setting up xfonts-utils (1:7.7+6build2) …
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.3) …
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) …
Setting up texlive-base (2021.20220204-1) …
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST…
```

```
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN…
mktexlsr: Updating /var/lib/texmf/ls-R…
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) …
Setting up texlive-plain-generic (2021.20220204-1) …
Setting up texlive-latex-base (2021.20220204-1) …
Setting up texlive-latex-recommended (2021.20220204-1) …
Setting up texlive-pictures (2021.20220204-1) …
Setting up texlive-fonts-recommended (2021.20220204-1) …
Setting up tipa (2:1.3-21) …
Setting up texlive-latex-extra (2021.20220204-1) …
Setting up texlive-xetex (2021.20220204-1) …
Setting up rake (13.0.6-2) …
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.12) …
Setting up ruby3.0 (3.0.2-7ubuntu2.12) …
Setting up ruby (1:3.0~exp1) …
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) …
Processing triggers for man-db (2.10.2-1) …
Processing triggers for mailcap (3.70+nmu1ubuntu1) …
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) …
Processing triggers for libc-bin (2.35-0ubuntu3.13) …
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link
```

```
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

Processing triggers for tex-common (6.17) …
Running updmap-sys. This may take some time… done.
Running mktexlsr /var/lib/texmf … done.
Building format(s) --all.
        This may take some time… done.
[NbConvertApp] WARNING | pattern '/content/Assignment5.ipynb' matched no files
This application is used to convert notebook files (*.ipynb)
        to various other formats.

        WARNING: THE COMMANDLINE INTERFACE MAY CHANGE IN FUTURE RELEASES.

Options
=======
The options below are convenience aliases to configurable class-options,
as listed in the "Equivalent to" description-line of the aliases.
To see all configurable class-options for some <cmd>, use:
    <cmd> --help-all

--debug
    set log level to logging.DEBUG (maximize logging output)
    Equivalent to: [--Application.log_level=10]
--show-config
    Show the application's configuration (human-readable format)
    Equivalent to: [--Application.show_config=True]
--show-config-json
    Show the application's configuration (json format)
    Equivalent to: [--Application.show_config_json=True]
--generate-config
    generate default config file
    Equivalent to: [--JupyterApp.generate_config=True]
-y
    Answer yes to any questions instead of prompting.
    Equivalent to: [--JupyterApp.answer_yes=True]
--execute
    Execute the notebook prior to export.
    Equivalent to: [--ExecutePreprocessor.enabled=True]
```

```
--allow-errors
    Continue notebook execution even if one of the cells throws an error and
include the error message in the cell output (the default behaviour is to abort
conversion). This flag is only relevant if '--execute' was specified, too.
    Equivalent to: [--ExecutePreprocessor.allow_errors=True]
--stdin
    read a single notebook file from stdin. Write the resulting notebook with
default basename 'notebook.*'
    Equivalent to: [--NbConvertApp.from_stdin=True]
--stdout
    Write notebook output to stdout instead of files.
    Equivalent to: [--NbConvertApp.writer_class=StdoutWriter]
--inplace
    Run nbconvert in place, overwriting the existing notebook (only
            relevant when converting to notebook format)
    Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=]
--clear-output
    Clear output of current file and save in place,
            overwriting the existing notebook.
    Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=
--ClearOutputPreprocessor.enabled=True]
--coalesce-streams
    Coalesce consecutive stdout and stderr outputs into one stream (within each
cell).
    Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=
--CoalesceStreamsPreprocessor.enabled=True]
--no-prompt
    Exclude input and output prompts from converted document.
    Equivalent to: [--TemplateExporter.exclude_input_prompt=True
--TemplateExporter.exclude_output_prompt=True]
--no-input
    Exclude input cells and output prompts from converted document.
            This mode is ideal for generating code-free reports.
    Equivalent to: [--TemplateExporter.exclude_output_prompt=True
--TemplateExporter.exclude_input=True
--TemplateExporter.exclude_input_prompt=True]
--allow-chromium-download
    Whether to allow downloading chromium if no suitable version is found on the
system.
    Equivalent to: [--WebPDFExporter.allow_chromium_download=True]
--disable-chromium-sandbox
    Disable chromium security sandbox when converting to PDF..
    Equivalent to: [--WebPDFExporter.disable_sandbox=True]
--show-input
    Shows code input. This flag is only useful for dejavu users.
```

```
    Equivalent to: [--TemplateExporter.exclude_input=False]
--embed-images
    Embed the images as base64 dataurls in the output. This flag is only useful
for the HTML/WebPDF/Slides exports.
    Equivalent to: [--HTMLExporter.embed_images=True]
--sanitize-html
    Whether the HTML in Markdown cells and cell outputs should be sanitized..
    Equivalent to: [--HTMLExporter.sanitize_html=True]
--log-level=<Enum>
    Set the log level by value or name.
    Choices: any of [0, 10, 20, 30, 40, 50, 'DEBUG', 'INFO', 'WARN', 'ERROR',
'CRITICAL']
    Default: 30
    Equivalent to: [--Application.log_level]
--config=<Unicode>
    Full path of a config file.
    Default: ''
    Equivalent to: [--JupyterApp.config_file]
--to=<Unicode>
    The export format to be used, either one of the built-in formats
            ['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook',
'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf']
            or a dotted object name that represents the import path for an
            ``Exporter`` class
    Default: ''
    Equivalent to: [--NbConvertApp.export_format]
--template=<Unicode>
    Name of the template to use
    Default: ''
    Equivalent to: [--TemplateExporter.template_name]
--template-file=<Unicode>
    Name of the template file to use
    Default: None
    Equivalent to: [--TemplateExporter.template_file]
--theme=<Unicode>
    Template specific theme(e.g. the name of a JupyterLab CSS theme distributed
    as prebuilt extension for the lab template)
    Default: 'light'
    Equivalent to: [--HTMLExporter.theme]
--sanitize_html=<Bool>
    Whether the HTML in Markdown cells and cell outputs should be sanitized.This
    should be set to True by nbviewer or similar tools.
    Default: False
    Equivalent to: [--HTMLExporter.sanitize_html]
--writer=<DottedObjectName>
    Writer class used to write the
                                        results of the conversion
    Default: 'FilesWriter'
```

```
    Equivalent to: [--NbConvertApp.writer_class]
--post=<DottedOrNone>
    PostProcessor class used to write the
                                        results of the conversion
    Default: ''
    Equivalent to: [--NbConvertApp.postprocessor_class]
--output=<Unicode>
    Overwrite base name use for output files.
                Supports pattern replacements '{notebook_name}'.
    Default: '{notebook_name}'
    Equivalent to: [--NbConvertApp.output_base]
--output-dir=<Unicode>
    Directory to write output(s) to. Defaults
                                  to output to the directory of each notebook.
To recover
                                  previous default behaviour (outputting to the
current
                                  working directory) use . as the flag value.
    Default: ''
    Equivalent to: [--FilesWriter.build_directory]
--reveal-prefix=<Unicode>
    The URL prefix for reveal.js (version 3.x).
            This defaults to the reveal CDN, but can be any url pointing to a
copy
            of reveal.js.
            For speaker notes to work, this must be a relative path to a local
            copy of reveal.js: e.g., "reveal.js".
            If a relative path is given, it must be a subdirectory of the
            current directory (from which the server is run).
            See the usage documentation
            (https://nbconvert.readthedocs.io/en/latest/usage.html#reveal-js-
html-slideshow)
            for more details.
    Default: ''
    Equivalent to: [--SlidesExporter.reveal_url_prefix]
--nbformat=<Enum>
    The nbformat version to write.
            Use this to downgrade notebooks.
    Choices: any of [1, 2, 3, 4]
    Default: 4
    Equivalent to: [--NotebookExporter.nbformat_version]

Examples
--------

    The simplest way to use nbconvert is

            > jupyter nbconvert mynotebook.ipynb --to html
```

```
            Options include ['asciidoc', 'custom', 'html', 'latex', 'markdown',
'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides',
'webpdf'].

            > jupyter nbconvert --to latex mynotebook.ipynb

            Both HTML and LaTeX support multiple output templates. LaTeX
includes
            'base', 'article' and 'report'.  HTML includes 'basic', 'lab' and
            'classic'. You can specify the flavor of the format used.

            > jupyter nbconvert --to html --template lab mynotebook.ipynb

            You can also pipe the output to stdout, rather than a file

            > jupyter nbconvert mynotebook.ipynb --stdout

            PDF is generated via latex

            > jupyter nbconvert mynotebook.ipynb --to pdf

            You can get (and serve) a Reveal.js-powered slideshow

            > jupyter nbconvert myslides.ipynb --to slides --post serve

            Multiple notebooks can be given at the command line in a couple of
            different ways:

            > jupyter nbconvert notebook*.ipynb
            > jupyter nbconvert notebook1.ipynb notebook2.ipynb

            or you can specify the notebooks list in a config file, containing::

                c.NbConvertApp.notebooks = ["my_notebook.ipynb"]

            > jupyter nbconvert --config mycfg.py

To see all available configurables, use `--help-all`.
```

[18]:
```
!jupyter nbconvert --to pdf /content/drive/MyDrive/Colab\ Notebooks/Assignment5.
↪ipynb
```

```
[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab
Notebooks/Assignment5.ipynb to pdf
[NbConvertApp] Support files will be in Assignment5_files/
[NbConvertApp] Making directory ./Assignment5_files
```

```
[NbConvertApp] Writing 158274 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 922487 bytes to /content/drive/MyDrive/Colab
Notebooks/Assignment5.pdf
```

[16]:
```
!ls /content/
```

```
drive  sample_data
```

[ ]:
```

```